

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 12 • Integration

With Answers

Questions with full worked solutions

- 67 pages
- Past papers up to January 2026

PURE 2 | TOPIC 12

Integration

Your classified route

29

QUESTIONS

212

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 12 | FAMILY 01

Exact Area

WHAT THIS FAMILY ASKS

Recognise the exact area demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**9 classified questions**

78 total marks

T12

12_AREA

12_AREA_CORE

CHAPTER
Integration

FAMILY
Exact area

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core family move

9 marks

WMA12/01 OCTOBER 2019

Question 8

Past-paper question

WMA12-01_2019_Oct_Q08

8. Solutions relying on calculator technology are not acceptable in this question.

(i)

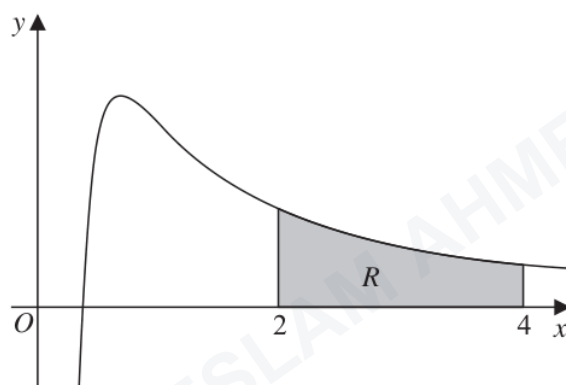


Figure 2

Figure 2 shows a sketch of part of a curve with equation

$$y = \frac{8\sqrt{x} - 5}{2x^2} \quad x > 0$$

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 4$

Find the exact area of R .

(5)

(ii) Find the value of the constant k such that

$$\int_{-3}^6 \left(\frac{1}{2}x^2 + k \right) dx = 55$$

(4)

T12 Integration

12_AREA Exact area

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move

WORKED SOLUTION

Question 8

WMA12/01 October 2019

WMA12-01_2019_Oct_Q08

1

Rewrite the curve

$$y = \frac{8\sqrt{x}-5}{2x^2} = 4x^{-3/2} - \frac{5}{2}x^{-2}.$$

2

Integrate the curve

$$\int \left(4x^{-3/2} - \frac{5}{2}x^{-2}\right) dx = -8x^{-1/2} + \frac{5}{2x}.$$

3

Apply the limits for the area

$$\left[-8x^{-1/2} + \frac{5}{2x}\right]_2^4 = \left(-4 + \frac{5}{8}\right) - \left(-4\sqrt{2} + \frac{5}{4}\right).$$

4

Simplify the exact area

$$\text{This gives } 4\sqrt{2} - \frac{37}{8}.$$

5

Integrate the second expression

$$\int \left(\frac{1}{2}x^2 + k\right) dx = \frac{x^3}{6} + kx.$$

6

Use the limits

$$\left[\frac{x^3}{6} + kx\right]_{-3}^6 = (36 + 6k) - \left(-\frac{9}{2} - 3k\right) = \frac{81}{2} + 9k.$$

7

Solve for k

$$\frac{81}{2} + 9k = 55, \text{ so } 9k = \frac{29}{2} \text{ and } k = \frac{29}{18}.$$

FINAL ANSWER

$$(i) 4\sqrt{2} - \frac{37}{8}. \quad (ii) k = \frac{29}{18}.$$

WMA12-01_2019_Oct_Q08

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6 marks

WMA12/01 JANUARY 2020

Question 4

Past-paper question

WMA12-01_2020_Jan_Q04

4.

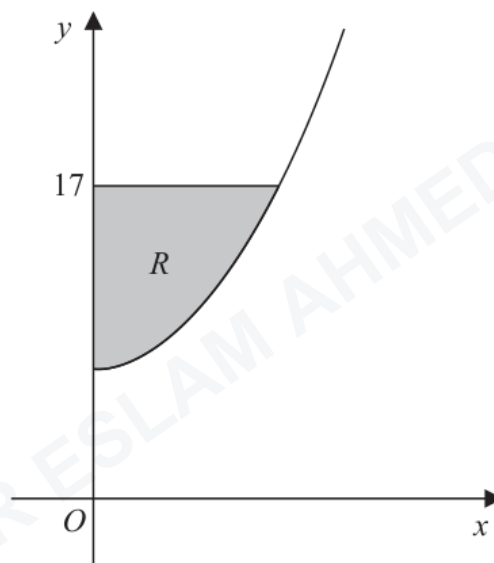


Figure 1

Figure 1 shows a sketch of the curve with equation

$$y = 2x^2 + 7 \quad x \geq 0$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the y -axis and the line with equation $y = 17$

Find the exact area of R .

(6)

T12 Integration

12_AREA Exact area

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WORKED SOLUTION

Question 4

WMA12/01 January 2020

WMA12-01_2020_Jan_Q04

1

Find the intersection with y equals 17

The curve is $y = 2x^2 + 7$. Set $2x^2 + 7 = 17$ to get $x^2 = 5$, so the right-hand limit is $x = \sqrt{5}$.

2

Set up the area

The shaded area is between $y = 17$ and $y = 2x^2 + 7$, from $x = 0$ to $x = \sqrt{5}$.

3

Integrate

$$A = \int_0^{\sqrt{5}} \{17 - (2x^2 + 7)\} dx = \int_0^{\sqrt{5}} (10 - 2x^2) dx.$$

4

Evaluate the exact area

$$A = \left[10x - \frac{2}{3}x^3 \right]_0^{\sqrt{5}} = 10\sqrt{5} - \frac{2}{3}(5\sqrt{5}) = \frac{20\sqrt{5}}{3}.$$

FINAL ANSWER

$$\frac{20\sqrt{5}}{3}.$$

WMA12-01_2020_Jan_Q04

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10 marks

WMA12/01 JANUARY 2022

Question 9

Past-paper question

WMA12-01_2022_Jan_Q09

9.

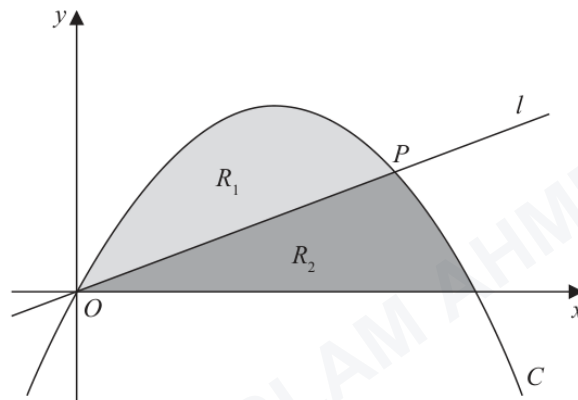


Figure 2

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .

(a) Find, in terms of m , the coordinates of P .

(2)

The region R_1 , shown shaded in Figure 2, is bounded by C and l .

(b) Show that the area of R_1 is

$$\frac{(1-m)^3}{6}$$

(5)

The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and l .

Given that the area of R_1 is equal to the area of R_2

(c) find the exact value of m .

(3)

T12 Integration

12_AREA Exact area

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WORKED SOLUTION

Question 9

WMA12/01 January 2022

WMA12-01_2022_Jan_Q09

1

Find the second intersectionAt an intersection, $x - x^2 = mx$, so $x(1 - m - x) = 0$.

2

State PThe non-zero intersection has $x = 1 - m$, and then $y = m(1 - m)$, so $P = (1 - m, m(1 - m))$.

3

Set up area R oneRegion R_1 is between the curve and the line from $x = 0$ to $x = 1 - m$.

4

Integrate the difference

$$A_{R_1} = \int_0^{1-m} \{(x - x^2) - mx\} dx = \int_0^{1-m} \{(1 - m)x - x^2\} dx.$$

5

Evaluate

$$A_{R_1} = \left[\frac{1-m}{2} x^2 - \frac{1}{3} x^3 \right]_0^{1-m} = \frac{(1-m)^3}{2} - \frac{(1-m)^3}{3} = \frac{(1-m)^3}{6}.$$

6

Use the total curve areaThe total area under $y = x - x^2$ from $x = 0$ to $x = 1$ is $\int_0^1 (x - x^2) dx = \frac{1}{6}$.

7

Use equal regionsSince R_1 and R_2 are equal and together make this total area, each has area $\frac{1}{12}$.

8

Solve for m

$$\frac{(1-m)^3}{6} = \frac{1}{12}, \text{ so } (1 - m)^3 = \frac{1}{2} \text{ and } m = 1 - \sqrt[3]{\frac{1}{2}}.$$

FINAL ANSWER

WMA12-01_2022_Jan_Q09

(a) $P = (1 - m, m(1 - m))$. (b) **Proven:** $\frac{(1-m)^3}{6}$. (c) $m = 1 - \sqrt[3]{\frac{1}{2}}$.

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8 marks

WMA12/01 JANUARY 2023
Question 9

Past-paper question

WMA12-01_2023_Jan_Q09

9. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

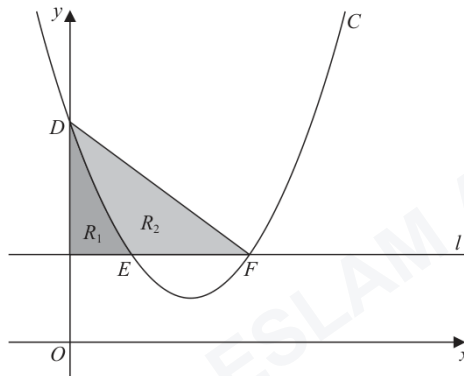


Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

(a) Write down the coordinates of D .

(1)

The curve C intersects the line l at the points E and F , as shown in Figure 3.

(b) Find the x coordinate of E and the x coordinate of F .

(2)

Shown shaded in Figure 3 is

- the region R_1 which is bounded by C , l and the y -axis
- the region R_2 which is bounded by C and the line segments EF and DF

Given that $\frac{\text{area of } R_1}{\text{area of } R_2} = k$, where k is a constant,

(c) use algebraic integration to find the exact value of k , giving your answer as a simplified fraction.

(5)

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WORKED SOLUTION

Question 9

WMA12/01 January 2023

WMA12-01_2023_Jan_Q09

1

Find DAt the y -axis, $x = 0$. On C , $y = 0^2 - 4(0) + 5 = 5$, so $D = (0, 5)$.

2

Find intersections with the lineThe line is $y = 2$, so solve $x^2 - 4x + 5 = 2$.

3

Solve the quadratic $x^2 - 4x + 3 = 0$, so $(x - 1)(x - 3) = 0$. Hence $x_E = 1$ and $x_F = 3$.

4

Area of R1 R_1 lies between the curve and the line $y = 2$ from $x = 0$ to $x = 1$.

5

Integrate R1

$$R_1 = \int_0^1 (x^2 - 4x + 5 - 2) dx = \int_0^1 (x^2 - 4x + 3) dx.$$

6

Evaluate R1

$$R_1 = \left[\frac{x^3}{3} - 2x^2 + 3x \right]_0^1 = \frac{1}{3} - 2 + 3 = \frac{4}{3}.$$

7

Find the line DFThe line through $D = (0, 5)$ and $F = (3, 2)$ has gradient -1 , so its equation is $y = 5 - x$.

8

Split R2For $0 \leq x \leq 1$, R_2 is between $y = 5 - x$ and the curve. For $1 \leq x \leq 3$, it is between $y = 5 - x$ and $y = 2$.

9

Integrate R2

$$R_2 = \int_0^1 [(5 - x) - (x^2 - 4x + 5)] dx + \int_1^3 [(5 - x) - 2] dx.$$

10

Evaluate R2

$$R_2 = \int_0^1 (-x^2 + 3x) dx + \int_1^3 (3 - x) dx = \frac{7}{6} + 2 = \frac{19}{6}.$$

11

Find the ratio

$$k = \frac{\text{area of } R_1}{\text{area of } R_2} = \frac{4/3}{19/6} = \frac{8}{19}.$$

FINAL ANSWER

WMA12-01_2023_Jan_Q09

$$(a) D = (0, 5). \quad (b) x_E = 1, x_F = 3. \quad (c) k = \frac{8}{19}.$$

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10 marks

WMA12/01 MAY/JUNE 2023

Question 10

Past-paper question

WMA12-01_2023_MayJune_Q10

10. The curve C has equation

$$y = \frac{(x-k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

(a) Show that

$$\int_1^{16} \frac{(x-k)^2}{\sqrt{x}} \, dx = ak^2 + bk + \frac{2046}{5}$$

where a and b are integers to be found.

(5)

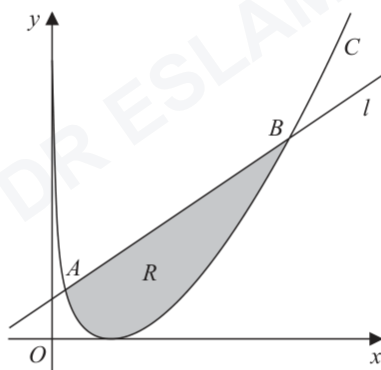


Figure 1

Figure 1 shows a sketch of the curve C and the line l .

Given that l intersects C at the point $A(1, 9)$ and at the point $B(16, q)$ where q is a constant,

(b) show that $k = 4$

(2)

The region R , shown shaded in Figure 1, is bounded by C and l

Using the answers to parts (a) and (b),

(c) find the area of region R

(3)

T12 Integration

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move

WORKED SOLUTION

Question 10

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q10

1

Expand the integrand

$$\frac{(x-k)^2}{\sqrt{x}} = \frac{x^2-2kx+k^2}{x^{1/2}} = x^{3/2} - 2kx^{1/2} + k^2x^{-1/2}.$$

2

Integrate

$$\int (x^{3/2} - 2kx^{1/2} + k^2x^{-1/2}) dx = \frac{2}{5}x^{5/2} - \frac{4k}{3}x^{3/2} + 2k^2x^{1/2}.$$

3

Substitute the limits

$$\text{From 1 to 16, this gives } \left(\frac{2048}{5} - \frac{256k}{3} + 8k^2\right) - \left(\frac{2}{5} - \frac{4k}{3} + 2k^2\right).$$

4

Simplify the integral

$$\text{The result is } 6k^2 - 84k + \frac{2046}{5}, \text{ so } a = 6 \text{ and } b = -84.$$

5

Use point A

$$\text{Since } A(1,9) \text{ lies on } C, 9 = \frac{(1-k)^2}{\sqrt{1}} = (1-k)^2.$$

6

Find k

$$\text{Thus } 1 - k = \pm 3, \text{ so } k = 4 \text{ or } k = -2. \text{ Since } k \text{ is positive, } k = 4.$$

7

Find point B

$$\text{When } k = 4 \text{ and } x = 16, q = \frac{(16-4)^2}{\sqrt{16}} = \frac{144}{4} = 36, \text{ so } B = (16, 36).$$

8

Find the line

$$\text{The line through } A(1,9) \text{ and } B(16,36) \text{ has gradient } \frac{36-9}{16-1} = \frac{9}{5}, \text{ so } l \text{ has equation } y = \frac{9}{5}x + \frac{36}{5}.$$

9

Area under the line

$$\int_1^{16} \left(\frac{9}{5}x + \frac{36}{5}\right) dx = \left[\frac{9}{10}x^2 + \frac{36}{5}x\right]_1^{16} = \frac{675}{2}.$$

10

Area under the curve

From part (a), with $k = 4$, $\int_1^{16} \frac{(x-4)^2}{\sqrt{x}} dx = 6(4)^2 - 84(4) + \frac{2046}{5} = \frac{846}{5}$.

11

Subtract curve from line

The shaded area is $\frac{675}{2} - \frac{846}{5} = \frac{1683}{10}$.

FINAL ANSWER

WMA12-01_2023_MayJune_Q10

(a) $a = 6$, $b = -84$. (b) $k = 4$. (c) **Area** = $\frac{1683}{10}$.

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8 marks

WMA12/01 OCTOBER 2024

Question 8

Past-paper question

WMA12-01_2024_Oct_Q08

8. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

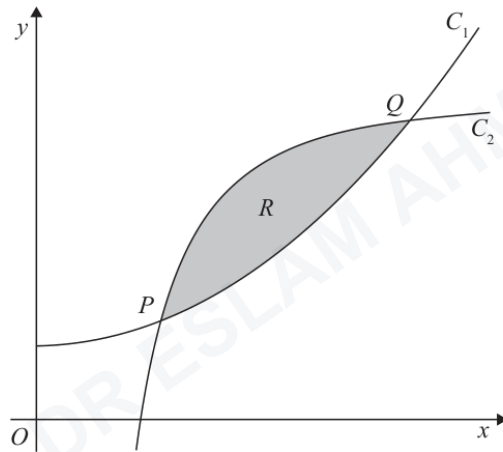


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation

$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.

- (a) Use algebra to find the x coordinate of P and the x coordinate of Q .

(4)

The finite region R , shown shaded in Figure 1, is bounded by C_1 and C_2

- (b) Use algebraic integration to find the exact area of R .

(4)

T12 Integration

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WORKED SOLUTION

Question 8

WMA12/01 October 2024

WMA12-01_2024_Oct_Q08

1

Equate the curves

$$x^2 + 3 = 13 - \frac{9}{x^2}.$$

2

Clear the denominator

Multiplying by x^2 gives $x^4 + 3x^2 = 13x^2 - 9$, so $x^4 - 10x^2 + 9 = 0$.

3

Factor

Let $u = x^2$. Then $u^2 - 10u + 9 = (u - 1)(u - 9) = 0$, so $x^2 = 1$ or $x^2 = 9$.

4

Use x greater than zero

Since $x > 0$, the intersections have $x = 1$ and $x = 3$. Thus $x_P = 1$ and $x_Q = 3$.

5

Set up the area

Between $x = 1$ and $x = 3$, curve C_2 is above curve C_1 , so $R = \int_1^3 \left(13 - \frac{9}{x^2} - (x^2 + 3)\right) dx$.

6

Simplify the integrand

$$R = \int_1^3 (10 - x^2 - 9x^{-2}) dx.$$

7

Integrate

$$R = \left[10x - \frac{x^3}{3} + \frac{9}{x}\right]_1^3.$$

8

Evaluate

$$R = (30 - 9 + 3) - \left(10 - \frac{1}{3} + 9\right) = 24 - \frac{56}{3} = \frac{16}{3}.$$

FINAL ANSWER

WMA12-01_2024_Oct_Q08

(a) $x_P = 1, x_Q = 3$. (b) $\frac{16}{3}$.

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12 marks

WMA12/01 JANUARY 2025

Question 9

Past-paper question

WMA12-01_2025_Jan_Q09

9.

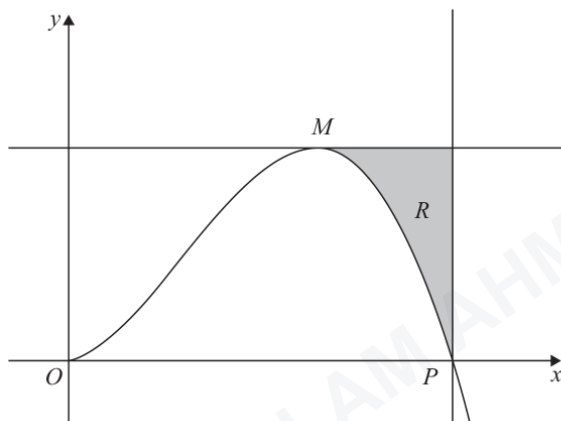


Figure 3

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

(a) Using calculus, find the coordinates of M .

(5)

The curve crosses the x -axis at the point P , as shown in Figure 3.

(b) Use algebra to find the x coordinate of P .

(2)

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line through M parallel to the x -axis and the line through P parallel to the y -axis.

(c) Use algebraic integration to find the area of R , giving your answer to one decimal place.

(5)

T12 Integration

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WORKED SOLUTION

Question 9

WMA12/01 January 2025

WMA12-01_2025_Jan_Q09

1

Rewrite the curve

$$y = \frac{9x^2(5-\sqrt{x})}{5} = 9x^2 - \frac{9}{5}x^{5/2}.$$

2

Differentiate

$$\frac{dy}{dx} = 18x - \frac{9}{2}x^{3/2} = \frac{9}{2}x(4 - \sqrt{x}).$$

3

Find the turning point x coordinate

For the turning point shown away from the origin, set $4 - \sqrt{x} = 0$. Hence $x = 16$.

4

Find the y coordinate

$$\text{At } x = 16, y = \frac{9(16)^2(5-4)}{5} = \frac{2304}{5}.$$

5

State M

Therefore $M\left(16, \frac{2304}{5}\right)$.

6

Find the positive x intercept

For the positive intercept, $\frac{9x^2(5-\sqrt{x})}{5} = 0$ gives $5 - \sqrt{x} = 0$, so $x = 25$.

7

Set up the area

The top boundary is $y = \frac{2304}{5}$ and the curve is below it from $x = 16$ to $x = 25$. So

$$R = \int_{16}^{25} \left(\frac{2304}{5} - 9x^2 + \frac{9}{5}x^{5/2} \right) dx.$$

8

Integrate

$$R = \left[\frac{2304}{5}x - 3x^3 + \frac{18}{35}x^{7/2} \right]_{16}^{25}.$$

9

Evaluate

$R = \frac{9189}{7} = 1312.714 \dots$, so the area is 1312.7 to one decimal place.

FINAL ANSWER

(a) $M\left(16, \frac{2304}{5}\right)$. (b) $x_P = 25$. (c) 1312.7.

WMA12-01_2025_Jan_Q09

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5 marks

WMA12/01 MAY/JUNE 2025

Question 3

Past-paper question

WMA12-01_2025_MayJune_Q03

3:

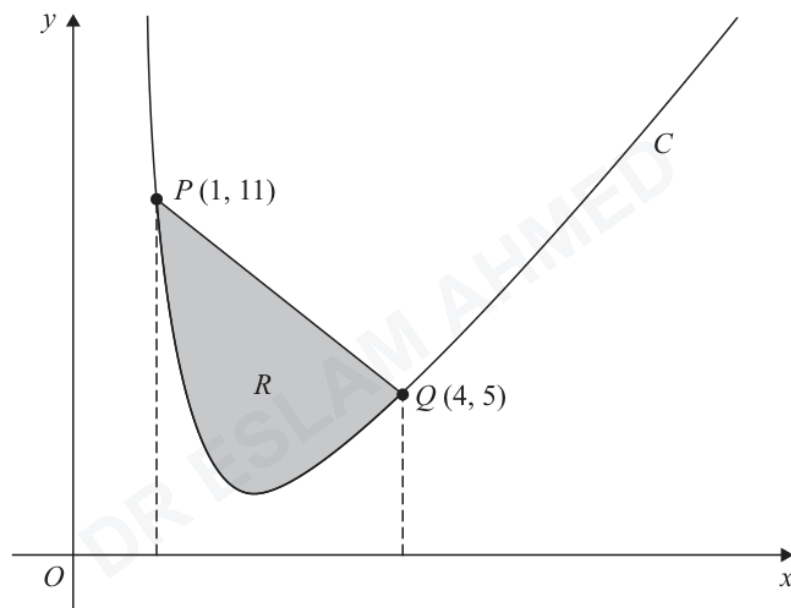


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve C with equation

$$y = 3x + \frac{16}{x^2} - 8 \quad x > 0$$

The points $P(1, 11)$ and $Q(4, 5)$ lie on C and are shown in Figure 1.

The region R , shown shaded in Figure 1, is bounded by C and line segment PQ .

Use algebraic integration to find the area of R .

(5)

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WORKED SOLUTION

Question 3

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q03

1

Find the line PQ

The gradient of PQ is $\frac{5-11}{4-1} = -2$, so the line through $P(1, 11)$ is $y - 11 = -2(x - 1)$.

2

Write the line equation

Thus PQ has equation $y = -2x + 13$.

3

Set up the area

From $x = 1$ to $x = 4$, the line segment is above the curve. Hence $R = \int_1^4 \left[(-2x + 13) - \left(3x + \frac{16}{x^2} - 8 \right) \right] dx$.

4

Simplify the integrand

$$R = \int_1^4 (21 - 5x - 16x^{-2}) dx.$$

5

Integrate

$$R = \left[21x - \frac{5}{2}x^2 + \frac{16}{x} \right]_1^4.$$

6

Evaluate

$$R = (84 - 40 + 4) - \left(21 - \frac{5}{2} + 16 \right) = 48 - \frac{69}{2} = \frac{27}{2}.$$

FINAL ANSWER

$$\frac{27}{2}.$$

WMA12-01_2025_MayJune_Q03

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10 marks

WMA12/01 OCTOBER 2025

Question 7

Past-paper question

WMA12-01_2025_Oct_Q07

7.

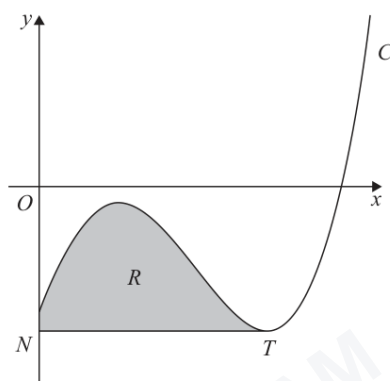


Figure 3

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

- (a) Find $\frac{dy}{dx}$ in simplest form. (2)

The point T , shown in Figure 3, is a minimum stationary point on C .

Given that the x coordinate of T is 9

- (b) show that $k = 6$ (2)

The line through T parallel to the x -axis meets the y -axis at the point N .

The finite region R , shown shaded in Figure 3, is bounded by C , the y -axis and the line segment NT .

- (c) Use algebraic integration to find the area of R , giving the answer to 3 significant figures. (6)

T12 Integration

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WORKED SOLUTION

Question 7

WMA12/01 October 2025

WMA12-01_2025_Oct_Q07

1

Differentiate

For $y = x^3 - 4x^{5/2} - kx^{1/2} + 28x - 44$, $\frac{dy}{dx} = 3x^2 - 10x^{3/2} - \frac{k}{2}x^{-1/2} + 28$.

2

Write in simplest form

$$\frac{dy}{dx} = 3x^2 - 10x^{3/2} - \frac{k}{2\sqrt{x}} + 28.$$

3

Use the stationary point

At T , $x = 9$ and $\frac{dy}{dx} = 0$.

4

Substitute x equals nine

$$0 = 3(9)^2 - 10(9)^{3/2} - \frac{k}{2\sqrt{9}} + 28 = 243 - 270 - \frac{k}{6} + 28.$$

5

Find k

This simplifies to $0 = 1 - \frac{k}{6}$, so $k = 6$.

6

Find the horizontal line

With $k = 6$, the y coordinate of T is $9^3 - 4(9)^{5/2} - 6\sqrt{9} + 28(9) - 44 = -53$.

7

Set up the area

The line NT is $y = -53$. The region is between the curve and this line from $x = 0$ to $x = 9$.

8

Write the integral

$$R = \int_0^9 (x^3 - 4x^{5/2} - 6x^{1/2} + 28x - 44 + 53) dx.$$

9

Integrate

$$R = \left[\frac{x^4}{4} - \frac{8}{7}x^{7/2} - 4x^{3/2} + 14x^2 + 9x \right]_0^9.$$

10

Evaluate

$R = \frac{6939}{28} = 247.821 \dots$, so the area is 248 to 3 significant figures.

FINAL ANSWER

WMA12-01_2025_Oct_Q07

(a) $3x^2 - 10x^{3/2} - \frac{k}{2\sqrt{x}} + 28$. (b) **Proven.** (c) 248.

DR ESLAM AHMED

TOPIC 12 | FAMILY 02

Power Integration

WHAT THIS FAMILY ASKS

Recognise the power integration demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

12 total marks

T12

12_POWER

12_POWER_CORE

CHAPTER
Integration

FAMILY
Power integration

VARIANT
core family move

7 marks

WMA12/01 MAY/JUNE 2022

Question 7

Past-paper question

WMA12-01_2022_MayJune_Q07

7. $f(x) = Ax^3 + 6x^2 - 4x + B$

where A and B are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$

- $\int_3^5 f(x) dx = 176$

find the value of A and the value of B .

(7)

T12 Integration

12_POWER Power integration

12_POWER_CORE core
family move

WORKED SOLUTION

Question 7

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q07

1

Use the factor conditionSince $(x + 2)$ is a factor, $f(-2) = 0$.

2

Form the first equation $A(-2)^3 + 6(-2)^2 - 4(-2) + B = 0$, so $-8A + 32 + B = 0$ and $B = 8A - 32$.

3

Set up the integral

$$\int_3^5 (Ax^3 + 6x^2 - 4x + B) dx = 176.$$

4

Integrate

$$\left[\frac{A}{4}x^4 + 2x^3 - 2x^2 + Bx \right]_3^5 = 176.$$

5

Simplify the second equationThis gives $136A + 164 + 2B = 176$, so $68A + B = 6$.

6

Solve simultaneouslyUsing $B = 8A - 32$ in $68A + B = 6$ gives $76A = 38$, so $A = \frac{1}{2}$.

7

Find B

$$B = 8\left(\frac{1}{2}\right) - 32 = -28.$$

FINAL ANSWER

$$A = \frac{1}{2}, \quad B = -28.$$

WMA12-01_2022_MayJune_Q07

T12

12_POWER

12_POWER_CORE

CHAPTER
Integration

FAMILY
Power integration

VARIANT
core family move

5 marks

WMA12/01 JANUARY 2026

Question 4

Past-paper question

WMA12-01_2026_Jan_Q04

4.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Given that k is a positive constant and

$$\int_k^{2k} \left(\frac{12}{x^2} + 4 \right) dx = 14$$

find the possible values of k .

(5)

T12 Integration

12_POWER Power integration

12_POWER_CORE core
family move

WORKED SOLUTION

Question 4

WMA12/01 January 2026

WMA12-01_2026_Jan_Q04

1

Integrate first

$$\int \left(\frac{12}{x^2} + 4 \right) dx = \int (12x^{-2} + 4) dx = -\frac{12}{x} + 4x.$$

2

Apply the limits

$$\left[-\frac{12}{x} + 4x \right]_k^{2k} = \left(-\frac{6}{k} + 8k \right) - \left(-\frac{12}{k} + 4k \right) = \frac{6}{k} + 4k.$$

3

Form a quadratic

Given the integral is 14, $\frac{6}{k} + 4k = 14$. Multiplying by k gives $4k^2 - 14k + 6 = 0$, so $2k^2 - 7k + 3 = 0$.

4

Solve for k

$(2k - 1)(k - 3) = 0$, hence $k = \frac{1}{2}$ or $k = 3$. Both are positive, so both are valid.

FINAL ANSWER

$$k = \frac{1}{2} \text{ or } k = 3.$$

WMA12-01_2026_Jan_Q04

TOPIC 12 | FAMILY 03

Transformed Estimates

WHAT THIS FAMILY ASKS

Recognise the transformed estimates demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 For $f(x+c)$ or $f(-x)$, check the limits by substitution.

V02 Do not round the known estimate too early.

8 classified questions

53 total marks

T12

12_SHIFT

12_SHIFT_HV01

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
For $\int f(x+c)$ or $\int f(-x)$, check
the limits by substitution.

6 marks

WMA12/01 JANUARY 2023
Question 1

Past-paper question

WMA12-01_2023_Jan_Q01

1.

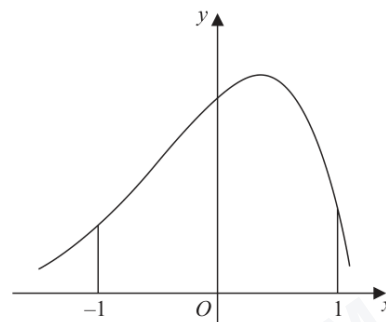


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$

The table below shows some corresponding values of x and y for this curve.

The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^1 f(x) \, dx$$

giving your answer to 2 decimal places.

(3)

(b) Use your answer to part (a) to estimate

(i) $\int_{-1}^1 (f(x) - 2) \, dx$

(ii) $\int_1^3 f(x-2) \, dx$

(3)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV01 For $f(x+c)$ or $f(-x)$, check the limits by substitution.

WORKED SOLUTION

Question 1

WMA12/01 January 2023

WMA12-01_2023_Jan_Q01

1

Use the trapezium widthThe x values run from -1 to 1 in steps of 0.5 , so $h = 0.5$.

2

Apply the trapezium rule

$$\int_{-1}^1 f(x) dx \approx \frac{0.5}{2} [2.287 + 2.834 + 2(4.470 + 6.719 + 7.291)].$$

3

Evaluate the estimateThis gives 10.52025 , so the estimate is 10.52 to 2 decimal places.

4

Estimate the shifted-down area

$$\int_{-1}^1 (f(x) - 2) dx = \int_{-1}^1 f(x) dx - \int_{-1}^1 2 dx.$$

5

Subtract the rectangleThe interval has width 2 , so $\int_{-1}^1 2 dx = 4$. Hence the estimate is $10.52 - 4 = 6.52$.

6

Use a substitutionFor $\int_1^3 f(x-2) dx$, let $u = x - 2$. When $x = 1$, $u = -1$, and when $x = 3$, $u = 1$.

7

Recognise the same integralSo $\int_1^3 f(x-2) dx = \int_{-1}^1 f(u) du$, which has the same estimate 10.52 .**FINAL ANSWER**

WMA12-01_2023_Jan_Q01

(a) 10.52 . (b)(i) 6.52 . (b)(ii) 10.52 .

T12

12_SHIFT

12_SHIFT_HV01

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
For $f(x+c)$ or $f(-x)$, check
the limits by substitution.

9 marks

WMA12/01 JANUARY 2024
Question 4

Past-paper question

WMA12-01_2024_Jan_Q04

4. (a) Sketch the curve with equation

$$y = a^{-x} + 4$$

where a is a constant and $a > 1$

On your sketch show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote to the curve.

(3)

x	-4	-1.5	1	3.5	6	8.5
y	13	6.280	4.577	4.146	4.037	4.009

The table above shows corresponding values of x and y for $y = 3^{-\frac{1}{2}x} + 4$

The values of y are given to four significant figures, as appropriate.

Using the trapezium rule with all the values of y in the table,

- (b) find an approximate value for

$$\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx$$

giving your answer to two significant figures.

(3)

- (c) Using the answer to part (b), find an approximate value for

(i) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} \right) dx$

(ii) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx + \int_{-8.5}^4 \left(3^{\frac{1}{2}x} + 4 \right) dx$

(3)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV01 For $f(x+c)/$
or $f(x-c)/$, check the limits by
substitution.

WORKED SOLUTION

Question 4

WMA12/01 January 2024

WMA12-01_2024_Jan_Q04

1

Find the y-interceptFor $y = a^{-x} + 4$, set $x = 0$: $y = a^0 + 4 = 5$, so the curve meets the y -axis at $(0, 5)$.

2

Find the asymptoteSince $a > 1$, $a^{-x} \rightarrow 0$ as $x \rightarrow \infty$, so the horizontal asymptote is $y = 4$.

3

Describe the sketchThe curve is decreasing, passes through $(0, 5)$, and approaches the line $y = 4$ from above.

4

Use the trapezium widthThe x values increase by 2.5, so $h = 2.5$.

5

Apply the trapezium rule

$$\int_{-4}^{8.5} (3^{-x/2} + 4) dx \approx \frac{2.5}{2} [13 + 4.009 + 2(6.280 + 4.577 + 4.146 + 4.037)].$$

6

Round part b

This gives 68.86125, so to two significant figures the estimate is 69.

7

Remove the constant part

$$\int_{-4}^{8.5} 4 dx = 4(12.5) = 50.$$

8

Estimate part c i

$$\text{Using the answer to part (b), } \int_{-4}^{8.5} 3^{-x/2} dx \approx 69 - 50 = 19.$$

9

Use symmetry by substitutionFor $\int_{-8.5}^4 (3^{x/2} + 4) dx$, let $u = -x$. This changes it to the same integral as part (b).

10

Estimate part c ii

Therefore the two integrals in part (c)(ii) are approximately equal, so the total is $69 + 69 = 138$.

FINAL ANSWER

WMA12-01_2024_Jan_Q04

(a) **y-intercept** $(0, 5)$ and **asymptote** $y = 4$. (b) 69. (c)(i) 19. (c)(ii) 138.

DR ESLAM AHMED

T12

12_SHIFT

12_SHIFT_HV01

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
For $\int f(x+c)/$ or $\int f(-x)/$, check
the limits by substitution.

5 marks

WMA12/01 MAY/JUNE (R) 2024

Question 2

Past-paper question

WMA12-01_2024_MayJuneR_Q02

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (a) find the value of the constant a to 3 significant figures.

(3)

- (b) Use the given answer of 19.3 to find an approximate value for

$$\int_{-4}^5 (2f(x) - 3) \, dx$$

(2)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV01 For $\int (f(x+c))$
or $\int (f(-x))$, check the limits by
substitution.

WORKED SOLUTION

Question 2

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q02

1

Use the trapezium widthThe x values increase by 1.5, so $h = 1.5$.

2

Set up the trapezium estimate

$$19.3 = \frac{1.5}{2} [4.16 + 2.28 + 2(2.91 + a + 1.73 + 1.37 + 1.43)].$$

3

Solve for a This simplifies to $19.3 = 15.99 + 1.5a$, so $a = \frac{3.31}{1.5} = 2.206 \dots$

4

Round a Therefore $a = 2.21$ to 3 significant figures.

5

Use the transformed integral

$$\int_{-4}^5 (2f(x) - 3) dx = 2 \int_{-4}^5 f(x) dx - \int_{-4}^5 3 dx.$$

6

Substitute the given estimateThis is $2(19.3) - 3(9) = 38.6 - 27 = 11.6$.**FINAL ANSWER**(a) $a = 2.21$ to 3 s.f. (b) 11.6.

WMA12-01_2024_MayJuneR_Q02

T12

12_SHIFT

12_SHIFT_HV01

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
For $f(x+c)$ or $f(-x)$, check
the limits by substitution.

5 marks

WMA12/01 OCTOBER 2024

Question 1

Past-paper question

WMA12-01_2024_Oct_Q01

1. A continuous curve has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

- (a) find an estimate for

$$\int_{0.5}^{5.5} f(x) \, dx$$

giving your answer to one decimal place.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

$$\int_{0.5}^{5.5} (f(x) + 4x) \, dx$$

(2)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV01 For $f(x+c)$ or $f(-x)$, check the limits by substitution.

WORKED SOLUTION

Question 1

WMA12/01 October 2024

WMA12-01_2024_Oct_Q01

1

Find the strip widthThe x values increase by 1.25, so the trapezium width is $h = 1.25$.

2

Apply the trapezium rule

$$\int_{0.5}^{5.5} f(x) dx \approx \frac{1.25}{2} \{3.479 + 5.182 + 2(6.101 + 7.448 + 6.823)\}.$$

3

Evaluate the estimate

This gives 30.878125, which is 30.9 to one decimal place.

4

Split the second integral

$$\int_{0.5}^{5.5} (f(x) + 4x) dx = \int_{0.5}^{5.5} f(x) dx + \int_{0.5}^{5.5} 4x dx.$$

5

Integrate $4x$

$$\int_{0.5}^{5.5} 4x dx = [2x^2]_{0.5}^{5.5} = 2(5.5)^2 - 2(0.5)^2 = 60.$$

6

Use part aUsing the estimate from part (a), the required value is $30.9 + 60 = 90.9$.**FINAL ANSWER**

(a) 30.9. (b) 90.9.

WMA12-01_2024_Oct_Q01

T12

12_SHIFT

12_SHIFT_HV02

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
Do not round the known
estimate too early.

9 marks

WMA12/01 MAY/JUNE 2024

Question 6

Past-paper question

WMA12-01_2024_MayJune_Q06

6. (a) Sketch the curve with equation

$$y = a^x + 4$$

where a is a positive constant greater than 1

On your sketch, show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote of the curve

(3)

x	2	2.3	2.6	2.9	3.2	3.5
y	0	0.3246	0.8629	1.6643	2.7896	4.3137

The table shows corresponding values of x and y for

$$y = 2^x - 2x$$

with the values of y given to 4 decimal places as appropriate.

Using the trapezium rule with all the values of y in the given table,

- (b) obtain an estimate for $\int_2^{3.5} (2^x - 2x) \, dx$, giving your answer to 2 decimal places.

(3)

- (c) Using your answer to part (b) and making your method clear, estimate

(i) $\int_2^{3.5} (2^x + 2x) \, dx$

(ii) $\int_2^{3.5} (2^{x+1} - 4x) \, dx$

(3)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV02 Do not round
the known estimate too early.

WORKED SOLUTION

Question 6

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q06

1

Sketch key featuresFor $y = a^x + 4$ with $a > 1$, the curve is increasing.

2

Find the y-interceptAt $x = 0$, $y = a^0 + 4 = 5$, so the y -intercept is $(0, 5)$.

3

Find the asymptoteAs $x \rightarrow -\infty$, $a^x \rightarrow 0$, so the horizontal asymptote is $y = 4$.

4

Use the trapezium widthThe x values increase by 0.3, so $h = 0.3$.

5

Apply the trapezium rule

$$\int_2^{3.5} (2^x - 2x) dx \approx \frac{0.3}{2} [0 + 4.3137 + 2(0.3246 + 0.8629 + 1.6643 + 2.7896)].$$

6

Round part b

This gives 2.339475, so the estimate is 2.34 to 2 decimal places.

7

Estimate part c i

$$2^x + 2x = (2^x - 2x) + 4x, \text{ so } \int_2^{3.5} (2^x + 2x) dx \approx 2.34 + \int_2^{3.5} 4x dx.$$

8

Add the extra integral

$$\int_2^{3.5} 4x dx = [2x^2]_2^{3.5} = 16.5, \text{ so the estimate is } 2.34 + 16.5 = 18.84.$$

9

Estimate part c ii

$$2^{x+1} - 4x = 2(2^x - 2x), \text{ so the estimate is } 2(2.34) = 4.68.$$

FINAL ANSWER

WMA12-01_2024_MayJune_Q06

(a) y -intercept $(0, 5)$ and asymptote $y = 4$. (b) 2.34. (c) (i) 18.84. (c) (ii) 4.68.

T12

12_SHIFT

12_SHIFT_HV02

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
Do not round the known
estimate too early.

6 marks

WMA12/01 MAY/JUNE 2025

Question 5

Past-paper question

WMA12-01_2025_MayJune_Q05

x	-2	-0.5	1	2.5	4	5.5	7
y	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of x and y for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of y are given to 3 decimal places as appropriate.

- (a) Using the trapezium rule with all the values of y in the given table, obtain an estimate for

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x dx$$

giving the answer to one decimal place.

(3)

Using the answer to part (a) and making your method clear, estimate

(b) (i) $\int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} dx$

(ii) $\int_{-2}^7 (2^{-x} + 2x) dx$

(3)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV02 Do not round
the known estimate too early.

WORKED SOLUTION

Question 5

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q05

1

Find the strip widthThe x values increase by 1.5, so $h = 1.5$.

2

Apply the trapezium rule

$$\int_{-2}^7 3 \left(\frac{1}{2}\right)^x dx \approx \frac{1.5}{2} \{12 + 0.023 + 2(4.243 + 1.5 + 0.530 + 0.188 + 0.066)\}.$$

3

Evaluate

This gives 18.80775, which is 18.8 to one decimal place.

4

Use the power shift

$$3 \left(\frac{1}{2}\right)^{x+2} = \frac{1}{4} \cdot 3 \left(\frac{1}{2}\right)^x.$$

5

Estimate part b i

$$\text{So } \int_{-2}^7 3 \left(\frac{1}{2}\right)^{x+2} dx \approx \frac{1}{4} (18.80775) = 4.7019375, \text{ which is } 4.7.$$

6

Rewrite part b ii

$$\text{Since } 3 \left(\frac{1}{2}\right)^x = 3 \cdot 2^{-x}, \text{ we have } 2^{-x} = \frac{1}{3} \cdot 3 \left(\frac{1}{2}\right)^x.$$

7

Estimate part b ii

$$\int_{-2}^7 (2^{-x} + 2x) dx \approx \frac{18.80775}{3} + [x^2]_{-2}^7 = 6.26925 + 45 = 51.26925, \text{ which is } 51.3.$$

FINAL ANSWER

WMA12-01_2025_MayJune_Q05

(a) 18.8. (b)(i) 4.7. (b)(ii) 51.3.

T12

12_SHIFT

12_SHIFT_HV02

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
Do not round the known
estimate too early.

7 marks

WMA12/01 OCTOBER 2025
Question 3

Past-paper question

WMA12-01_2025_Oct_Q03

3.

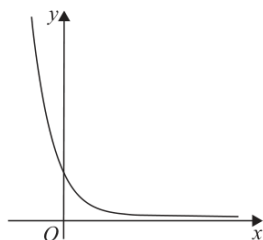


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation $y = 3 \times 2^{-x}$

The point $P(k, 300\,000)$ lies on the curve.

(a) Use logarithms to find the value of k to 2 decimal places.

(2)

x	-0.5	1	2.5	4.0	5.5	7
y	4.243	1.5	0.530	0.188	0.066	0.023

The table shows corresponding values of x and y for $y = 3 \times 2^{-x}$

The values of y are given to 3 decimal places where appropriate.

(b) (i) Use the trapezium rule, with all the values of y from the table, to find an approximate value, to 2 decimal places, for

$$\int_{-0.5}^7 3 \times 2^{-x} \, dx$$

(3)

(ii) Use your answer to part (b)(i) to estimate

$$\int_{-0.5}^7 2^{-x} \, dx + \int_{-7}^{0.5} 2^x \, dx$$

(2)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV02 Do not round
the known estimate too early.

WORKED SOLUTION

Question 3

WMA12/01 October 2025

WMA12-01_2025_Oct_Q03

1

Use the point on the curve

Since $P(k, 300000)$ lies on $y = 3 \times 2^{-x}$, $300000 = 3 \times 2^{-k}$.

2

Solve with logarithms

$100000 = 2^{-k}$, so $-k = \log_2(100000)$ and $k = -\log_2(100000) = -16.609 \dots$

3

Round k

To 2 decimal places, $k = -16.61$.

4

Find the strip width

The x values increase by 1.5, so $h = 1.5$.

5

Apply the trapezium rule

$$\int_{-0.5}^7 3 \times 2^{-x} dx \approx \frac{1.5}{2} \{4.243 + 0.023 + 2(1.5 + 0.530 + 0.188 + 0.066)\}.$$

6

Evaluate the estimate

This gives 6.6255, which is 6.63 to 2 decimal places.

7

Relate the first new integral

$$\int_{-0.5}^7 2^{-x} dx = \frac{1}{3} \int_{-0.5}^7 3 \times 2^{-x} dx.$$

8

Relate the second new integral

Using the substitution $u = -x$, $\int_{-7}^{0.5} 2^x dx = \int_{-0.5}^7 2^{-u} du$.

9

Estimate the total

The required estimate is therefore $\frac{2}{3}(6.6255) = 4.417$, which is 4.42 to 2 decimal places.

FINAL ANSWER

WMA12-01_2025_Oct_Q03

(a) $k = -16.61$. (b)(i) 6.63. (b)(ii) 4.42.

T12

12_SHIFT

12_SHIFT_HV02

CHAPTER
Integration

FAMILY
Transformed estimates

VARIANT
Do not round the known
estimate too early.

6 marks

WMA12/01 JANUARY 2026
Question 2

Past-paper question

WMA12-01_2026_Jan_Q02

2.

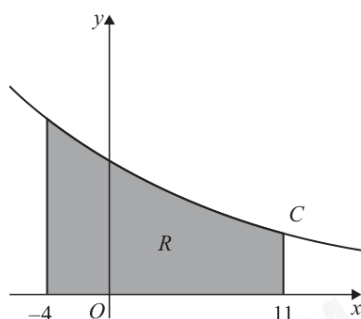


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = 2^{-0.1x}$

The table below shows corresponding values of x and y for $y = 2^{-0.1x}$

The y values are given to 4 decimal places.

x	-4	-1	2	5	8	11
y	1.3195	1.0718	0.8706	0.7071	0.5743	0.4665

The region R , shown shaded in Figure 1, is bounded by C , the x -axis and the lines with equations $x = -4$ and $x = 11$

(a) Use the trapezium rule with all the values of y in the table to find an estimate for the area of R . Give the answer to 2 decimal places.

(3)

(b) State how you would use the trapezium rule to get a more accurate estimate for the true area of R .

(1)

Using the answer to part (a) and showing your working,

(c) estimate the value of

$$\int_{-4}^{11} 2^{3-0.1x} dx$$

(2)

T12 Integration

12_SHIFT Transformed
estimates12_SHIFT_HV02 Do not round
the known estimate too early.

WORKED SOLUTION

Question 2

WMA12/01 January 2026

WMA12-01_2026_Jan_Q02

1

Find the strip widthThe x values increase by 3, so $h = 3$.

2

Apply the trapezium rule

$$A \approx \frac{3}{2} \{1.3195 + 0.4665 + 2(1.0718 + 0.8706 + 0.7071 + 0.5743)\}.$$

3

Calculate the estimated area

$$A \approx \frac{3}{2} (8.2336) = 12.3504, \text{ so the area is } 12.35 \text{ to 2 decimal places.}$$

4

Improve the estimate

A more accurate estimate would use more trapezia, so the strip width is smaller.

5

Use the scale factor in the integral

$$\text{Since } 2^{3-0.1x} = 8 \cdot 2^{-0.1x}, \text{ the required estimate is } 8(12.35) = 98.8.$$

FINAL ANSWER

WMA12-01_2026_Jan_Q02

(a) 12.35. (b) **Use narrower strips, or more trapezia.** (c) 98.8.

TOPIC 12 | FAMILY 04

Numerical Integration

WHAT THIS FAMILY ASKS

Recognise the numerical integration demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

10 classified questions

69 total marks

T12

12_TRAP

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CHAPTER
Integration

FAMILY
Numerical integration

VARIANT
core family move

7 marks

WMA12/01 OCTOBER 2019

Question 5

Past-paper question

WMA12-01_2019_Oct_Q05

5. (a) Given $0 < a < 1$, sketch the curve with equation

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y-axis.

(2)

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

- (b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x \right) dx$

(3)

Using your answer to part (b) and making your method clear, estimate

- (c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) dx$

(2)

T12 Integration

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move

WORKED SOLUTION

Question 5

WMA12/01 October 2019

WMA12-01_2019_Oct_Q05

1

Describe the sketch

For $0 < a < 1$, $y = a^x$ is a decreasing exponential curve. At $x = 0$, $y = a^0 = 1$, so it crosses the y -axis at $(0, 1)$.

2

Find the strip width

The x values increase by 0.5, so $h = 0.5$.

3

Apply the trapezium rule

$$A \approx \frac{0.5}{2} \{4.25 + 16.06 + 2(6.427 + 9.125 + 12.34)\}.$$

4

Calculate the estimate

$A \approx 0.25(76.094) = 19.0235$, so the estimate is 19.0 to 3 significant figures.

5

Relate the two integrands

$$x(x - 3) + \left(\frac{1}{2}\right)^x = x^2 + \left(\frac{1}{2}\right)^x - 3x.$$

6

Subtract the exact integral of $3x$

$$\int_2^4 3x \, dx = \left[\frac{3}{2}x^2\right]_2^4 = 24 - 6 = 18.$$

7

Estimate the new integral

Using the estimate from part (b), the required value is $19.0235 - 18 = 1.0235$, so the estimate is 1.0.

FINAL ANSWER

WMA12-01_2019_Oct_Q05

(a) A decreasing exponential curve crossing the y -axis at $(0, 1)$.

(b) 19.0. (c) 1.0.

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7 marks

WMA12/01 JANUARY 2020

Question 1

Past-paper question

WMA12-01_2020_Jan_Q01

1. The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for $\int_2^{14} \log_2(2x) dx$, giving your answer to one decimal place. (3)

Using your answer to part (a) and making your method clear, estimate

- (b) (i) $\int_2^{14} \frac{\log_2(4x^2)}{5} dx$
- (ii) $\int_2^{14} \log_2\left(\frac{2}{x}\right) dx$ (4)

T12 Integration

12_TRAP Numerical
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WORKED SOLUTION

Question 1

WMA12/01 January 2020

WMA12-01_2020_Jan_Q01

1

Use the trapezium rule

The strip width is $h = 3$. Therefore $\int_2^{14} \log_2(2x) dx \approx \frac{3}{2} \{2 + 4.81 + 2(3.32 + 4 + 4.46)\}$.

2

Calculate part (a)

The estimate is $\frac{3}{2}(30.37) = 45.555$, so to one decimal place the answer is 45.6.

3

Rewrite part (b)(i)

$\log_2(4x^2) = \log_2\{(2x)^2\} = 2\log_2(2x)$, so $\int_2^{14} \frac{\log_2(4x^2)}{5} dx = \frac{2}{5} \int_2^{14} \log_2(2x) dx$.

4

Estimate part (b)(i)

Using part (a), the estimate is $\frac{2}{5}(45.555) = 18.222$, so 18.2 to one decimal place.

5

Rewrite part (b)(ii)

$\log_2\left(\frac{2}{x}\right) = 2 - \log_2(2x)$.

6

Estimate part (b)(ii)

$\int_2^{14} \log_2\left(\frac{2}{x}\right) dx \approx \int_2^{14} 2 dx - 45.555 = 24 - 45.555 = -21.555$, so -21.6.

FINAL ANSWER

WMA12-01_2020_Jan_Q01

(a) 45.6. (b)(i) 18.2. (b)(ii) -21.6.

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6 marks

WMA12/01 JANUARY 2021

Question 7

Past-paper question

WMA12-01_2021_Jan_Q07

7.

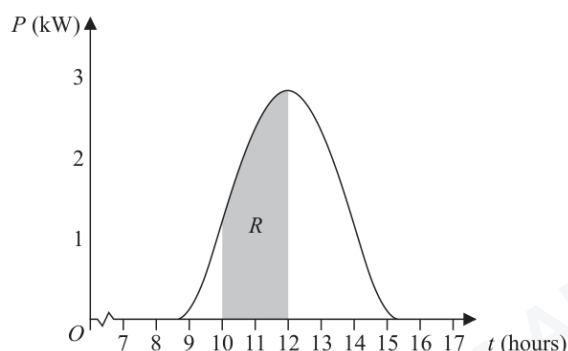


Figure 1

Solar panels are installed on the roof of a building.

The power, P , produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t-12)^2 \quad 8.5 \leq t \leq 15.2$$

where t is the time in hours after midnight. The graph of P against t is shown in Figure 1.

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

- (a) Use the given equation to complete the table, giving the values of P to 4 significant figures where appropriate.

(2)

The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region R , shown shaded in Figure 1.

- (b) Use the trapezium rule, with all the values of P in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.

(4)

T12 Integration

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move

WORKED SOLUTION

Question 7

WMA12/01 January 2021

WMA12-01_2021_Jan_Q07

1

Use the model at t equals ten

$$P(10) = 0.95 + 2^{-2} + 2^2 - (10 - 12)^2 = 0.95 + 0.25 + 4 - 4 = 1.20.$$

2

Use the model at t equals eleven point five

$$P(11.5) = 0.95 + 2^{-0.5} + 2^{0.5} - (11.5 - 12)^2 = 2.821 \dots$$

3

Identify the trapezium width

The time values increase by 0.5, so $h = 0.5$.

4

Apply the trapezium rule

The estimate is $\frac{0.5}{2} \{1.20 + 2.95 + 2(1.882 + 2.45 + 2.821 \dots)\}$.

5

Calculate the energy

This gives 4.614 ..., so the estimated energy is 4.61 kWh to 2 decimal places.

FINAL ANSWER

WMA12-01_2021_Jan_Q07

(a) $P(10) = 1.20$, $P(11.5) = 2.821$. (b) 4.61 kWh.

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10 marks

WMA12/01 MAY/JUNE 2021

Question 5

Past-paper question

WMA12-01_2021_MayJune_Q05

5.

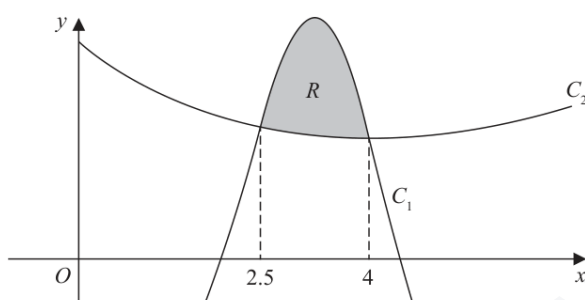


Figure 2

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

- (a) find, to 2 decimal places, an estimate for the area bounded by the curve C_1 , the line with equation $x = 2.5$, the x -axis and the line with equation $x = 4$

(4)

The curve C_2 has equation

$$y = x^{\frac{3}{2}} - 3x + 9 \quad x > 0$$

- (b) Find $\int \left(x^{\frac{3}{2}} - 3x + 9 \right) dx$

(3)

The region R , shown shaded in Figure 2, is bounded by the curves C_1 and C_2

- (c) Use the answers to part (a) and part (b) to find, to one decimal place, an estimate for the area of the region R .

(3)

T12 Integration

12_TRAP Numerical
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WORKED SOLUTION

Question 5

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q05

1

Identify the strip widthThe x values increase by 0.25, so $h = 0.25$.

2

Apply the trapezium ruleThe area under C_1 is approximately $\frac{0.25}{2}\{5.453 + 5 + 2(7.764 + 9.375 + 9.964 + 9.367 + 7.626)\}$.

3

Calculate the estimate

This gives 12.330625, so the estimate is 12.33 to 2 decimal places.

4

Integrate C two

$$\int (x^{3/2} - 3x + 9) dx = \frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x + c.$$

5

Find the area under C twoBetween $x = 2.5$ and $x = 4$, the exact integral of C_2 is $\left[\frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x\right]_{2.5}^4 = 7.722 \dots$

6

Find the shaded areaThe shaded region is between C_1 and C_2 , so its area is approximately $12.330625 - 7.722 \dots = 4.608 \dots$

7

Round the resultTo one decimal place, the area of region R is 4.6.**FINAL ANSWER**

WMA12-01_2021_MayJune_Q05

(a) 12.33. (b) $\frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x + c$. (c) 4.6.

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8 marks

WMA12/01 OCTOBER 2021

Question 3

Past-paper question

WMA12-01_2021_Oct_Q03

3.

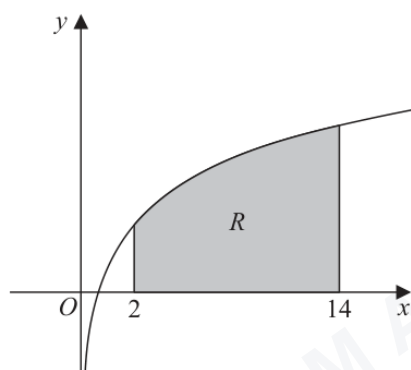


Figure 1

Figure 1 shows a sketch of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(a) show that the area of R is approximately 10.10 (3)

(b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R . (1)

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x} \, dx$

(ii) $\int_2^{14} \log_{10} 100x^3 \, dx$

(4)

T12 Integration

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move

WORKED SOLUTION

Question 3

WMA12/01 October 2021

WMA12-01_2021_Oct_Q03

1

Find the strip widthFour equal strips from $x = 2$ to $x = 14$ have width $h = \frac{14-2}{4} = 3$.

2

List the ordinatesUse $x = 2, 5, 8, 11, 14$ and $y = \log_{10} x$.

3

Apply the trapezium rule

$$A \approx \frac{3}{2} \{ \log_{10} 2 + \log_{10} 14 + 2(\log_{10} 5 + \log_{10} 8 + \log_{10} 11) \} = 10.10 \dots$$

4

Improve the estimate

Use more strips of smaller width to make the trapezium rule estimate more accurate.

5

Estimate the square-root integral

$$\log_{10} \sqrt{x} = \frac{1}{2} \log_{10} x, \text{ so } \int_2^{14} \log_{10} \sqrt{x} dx \approx \frac{1}{2} (10.10) = 5.05.$$

6

Rewrite the second logarithm

$$\log_{10}(100x^3) = \log_{10} 100 + \log_{10} x^3 = 2 + 3 \log_{10} x.$$

7

Estimate the second integral

$$\int_2^{14} (2 + 3 \log_{10} x) dx \approx 2(12) + 3(10.10) = 54.30.$$

FINAL ANSWER

WMA12-01_2021_Oct_Q03

(a) 10.10. (b) **Use more strips.** (c)(i) 5.05. (c)(ii) 54.30.

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7 marks

WMA12/01 JANUARY 2022

Question 1

Past-paper question

WMA12-01_2022_Jan_Q01

1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}} dx$$

giving your answer to 2 decimal places.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

(i) $\int_5^7 2^{6-\sqrt{x}} dx$

(ii) $\int_5^7 (3 + 2^{5-\sqrt{x}}) dx$

(4)

T12 Integration

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WORKED SOLUTION

Question 1

WMA12/01 January 2022

WMA12-01_2022_Jan_Q01

1

Identify the strip widthThe x values increase by 0.5, so $h = 0.5$.

2

Apply the trapezium rule

$$\int_5^7 2^{5-\sqrt{x}} dx \approx \frac{0.5}{2} \{6.792 + 5.113 + 2(6.298 + 5.858 + 5.466)\}.$$

3

Calculate the estimate

This gives 11.78725, so the estimate is 11.79 to 2 decimal places.

4

Use the factor of two

$$2^{6-\sqrt{x}} = 2 \cdot 2^{5-\sqrt{x}}, \text{ so } \int_5^7 2^{6-\sqrt{x}} dx \approx 2(11.78725) = 23.5745.$$

5

Estimate part i

This is 23.57 to 2 decimal places.

6

Add the constant integral

$$\int_5^7 (3 + 2^{5-\sqrt{x}}) dx = \int_5^7 3 dx + \int_5^7 2^{5-\sqrt{x}} dx.$$

7

Estimate part iiThe estimate is $3(7 - 5) + 11.78725 = 17.78725$, so 17.79 to 2 decimal places.**FINAL ANSWER**

WMA12-01_2022_Jan_Q01

(a) 11.79. (b)(i) 23.57. (b)(ii) 17.79.

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8 marks

WMA12/01 MAY/JUNE 2022

Question 2

Past-paper question

WMA12-01_2022_MayJune_Q02

2.

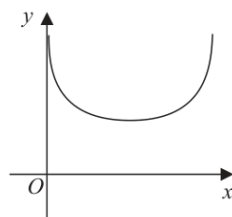


Figure 1

Figure 1 shows the graph of

$$y = 1 - \log_{10}(\sin x) \quad 0 < x < \pi$$

where x is in radians.

The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

- (a) Complete the table above, giving values of y to 3 decimal places.

(2)

- (b) Use the trapezium rule with all the y values in the completed table to find, to 2 decimal places, an estimate for

$$\int_{0.5}^3 (1 - \log_{10}(\sin x)) dx$$

(3)

- (c) Use your answer to part (b) to find an estimate for

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) dx$$

(3)

T12 Integration

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move

WORKED SOLUTION

Question 2

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q02

1

Complete the missing table valuesUsing $y = 1 - \log_{10}(\sin x)$ gives $y(1) = 1.075$ and $y(2) = 1.041$ to 3 decimal places.

2

Identify the strip widthThe x values increase by 0.5, so $h = 0.5$.

3

Apply the trapezium rule

$$A \approx \frac{0.5}{2} \{1.319 + 1.850 + 2(1.075 + 1.001 + 1.041 + 1.223)\}.$$

4

Calculate part b

This gives 2.9625 ..., so the estimate is 2.96 to 2 decimal places.

5

Rewrite the new integrandSince $y = 1 - \log_{10}(\sin x)$, we have $\log_{10}(\sin x) = 1 - y$.

6

Use the interval length $3 + \log_{10}(\sin x) = 4 - y$, and the interval from 0.5 to 3 has length 2.5.

7

Estimate part c

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) dx \approx 4(2.5) - 2.9625 \dots = 7.0374 \dots, \text{ so } 7.04.$$

FINAL ANSWER

WMA12-01_2022_MayJune_Q02

(a) $y(1) = 1.075$, $y(2) = 1.041$. (b) 2.96. (c) 7.04.

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7 marks

WMA12/01 OCTOBER 2022

Question 6

Past-paper question

WMA12-01_2022_Oct_Q06

6. The curve C_1 has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below, with the y values rounded to 4 decimal places where appropriate.

x	0	0.5	1	1.5	2
y	3	2.6833	2.4	2.1466	1.92

- (a) Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_0^2 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

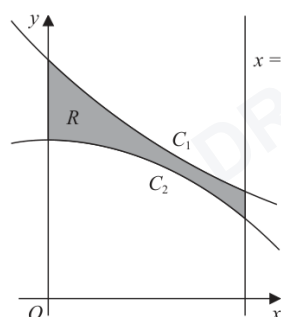


Figure 1

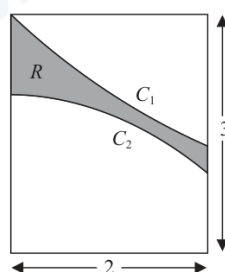


Figure 2

The region R , shown shaded in Figure 1, is bounded by

- the curve C_1
- the curve C_2 with equation $y = 2 - \frac{1}{4}x^2$
- the line with equation $x = 2$
- the y -axis

The region R forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region R inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

- (b) calculate an estimate for the percentage of the logo which is shaded.

(4)

T12 Integration

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WORKED SOLUTION

Question 6

WMA12/01 October 2022

WMA12-01_2022_Oct_Q06

1

Identify the strip widthThe x values increase by 0.5, so the trapezium-rule width is $h = 0.5$.

2

Apply the trapezium rule

$$\int_0^2 f(x) dx \approx \frac{0.5}{2} [3 + 1.92 + 2(2.6833 + 2.4 + 2.1466)].$$

3

Evaluate the estimate

This gives 4.84495, so the approximation is 4.845 to 3 decimal places.

4

Find the area under C2

$$\int_0^2 \left(2 - \frac{1}{4}x^2\right) dx = \left[2x - \frac{x^3}{12}\right]_0^2 = 4 - \frac{2}{3} = \frac{10}{3}.$$

5

Estimate the shaded areaThe shaded region lies between C_1 and C_2 , so its estimated area is $4.845 - \frac{10}{3} = 1.5116\dots$

6

Compare with the rectangleThe rectangle has area $2 \times 3 = 6$.

7

Convert to a percentagePercentage shaded = $\frac{1.5116\dots}{6} \times 100 = 25.19\dots\%$, so the estimate is 25.2%.**FINAL ANSWER**

(a) 4.845. (b) 25.2%.

WMA12-01_2022_Oct_Q06

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3 marks

WMA12/01 MAY/JUNE 2023

Question 1

Past-paper question

WMA12-01_2023_MayJune_Q01

1. The continuous curve C has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	4.0	4.2	4.4	4.6	4.8	5.0
y	9.2	8.4556	3.8512	5.0342	7.8297	8.6

Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_4^5 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

T12 Integration

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move

WORKED SOLUTION

Question 1

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q01

1

Identify the interval width

The x values go from 4.0 to 5.0 in steps of 0.2, so $h = 0.2$.

2

Apply the trapezium rule

$$\int_4^5 f(x) dx \approx \frac{0.2}{2} [9.2 + 8.6 + 2(8.4556 + 3.8512 + 5.0342 + 7.8297)].$$

3

Evaluate

The value is 6.81414, so the approximation is 6.814 to 3 decimal places.

FINAL ANSWER

6.814.

WMA12-01_2023_MayJune_Q01

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6 marks

WMA12/01 OCTOBER 2023

Question 6

Past-paper question

WMA12-01_2023_Oct_Q06

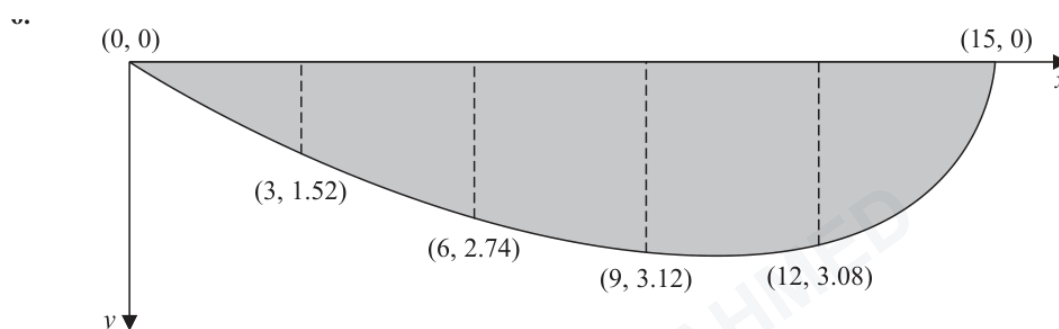


Figure 1

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

- (a) Use the trapezium rule with all the y values given in Figure 1 to estimate the cross-sectional area of the river.

(3)

The water in the river is modelled as flowing at a constant speed of 1.5 m s^{-1} across the whole of the cross-section.

- (b) Use the model and the answer to part (a) to estimate the volume of water flowing through this section of the river each minute, giving your answer in m^3 to 2 significant figures.

(2)

Assuming the model,

- (c) state, giving a reason for your answer, whether your answer for part (b) is an overestimate or an underestimate of the true volume of water flowing through this section of the river each minute.

(1)

T12 Integration

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move

WORKED SOLUTION

Question 6

WMA12/01 October 2023

WMA12-01_2023_Oct_Q06

1

Read the strip width

The x values are 0, 3, 6, 9, 12, 15, so the trapezium width is $h = 3$.

2

Apply the trapezium rule

Area $\approx \frac{3}{2} [0 + 0 + 2(1.52 + 2.74 + 3.12 + 3.08)]$.

3

Find the cross-sectional area

This gives 31.38 m^2 .

4

Convert flow speed to one minute

The speed is 1.5 m s^{-1} , so in one minute the water travels $1.5 \times 60 = 90 \text{ m}$.

5

Estimate volume

Volume per minute $\approx 31.38 \times 90 = 2824.2 \text{ m}^3$.

6

Round the volume

To 2 significant figures, this is $2.8 \times 10^3 \text{ m}^3$.

7

State over or under

It is an underestimate, because the curve between the plotted points lies below the straight-line trapezium chords in the diagram, so the true cross-sectional area is larger.

FINAL ANSWER

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(a) 31.38 m^2 . (b) $2.8 \times 10^3 \text{ m}^3$. (c) **Underestimate.**